

PRE-ALGEBRA — CLASS 1

Integers on the Number Line & Absolute Value

SOMATH

School of Math — NYC

Instructions. Read each question carefully. Show your work on the lines and write your final answer clearly. These are **25 challenging questions** that layer natural numbers, whole numbers, integers, absolute value, and divisibility rules — several will need two or three steps to solve. If you get stuck, sketch a number line and try again.

Name: _____

Date: _____

Score: _____ / 25

QUESTIONS 1-25

1. From the list -6, 0, 3, -1, 5, 0.5, -0.25, list every number that belongs to $\mathbb{Z}$ (the integers).

2. Which of these numbers are in $\mathbb{W}$ (the whole numbers) but NOT in $\mathbb{N}$ (the natural numbers)? List: 0, 1, 5, -2, 12, -7.

3. On a number line, point A is at -6 and point B is at 10. What integer is at the MIDPOINT of AB?

4. Order from LEAST to GREATEST: $-|-7|$, $-(-5)$, 0, $|-3|$, $-(4 - 9)$.

5. Fill each blank with $<$, $>$, or $=$: (a) $-(-8)$ ____ $|-8|$ (b) $-|-6|$ ____ $-(-6)$ (c) $-|4|$ ____ 0

6. At 6 a.m. the temperature was -12°F . It rose 3°F per hour for 5 hours. What was the temperature at 11 a.m.?

7. A submarine at -180 feet rises 25 feet per minute for 4 minutes. What is its new depth?

8. Maya's checking account started at $-\$35$ (overdrawn). She deposited $\$80$, then paid a $\$22$ bill. Write a signed-number expression for her ending balance and evaluate it.

9. Evaluate: $|-9| + |4 - 11| - |-2| \times |5|$.

10. Solve for x : (a) $|x| = 12$ (b) $|x - 4| = 7$. List every solution.

11. True or false, with a one-sentence reason: (a) $|a \times b| = |a| \times |b|$ for every pair of integers. (b) $|a|$ is always positive or zero, never negative.

12. List every integer x that satisfies $|x| < 4$.

13. A drone starts at 0 on a number line and moves: +6, -10, +4, -3. (a) Where does it end up? (b) What is the TOTAL DISTANCE it traveled?

14. Points P and Q are on the number line with $|P| = 7$ and $Q = -2$. Find every possible distance from P to Q.

15. Test each number for divisibility by 3 AND by 4: (a) 132 (b) 246 (c) 528.

16. Find the SMALLEST positive integer that is divisible by both 6 and 8.

17. Which of these numbers are divisible by 9? Numbers: 234, 405, 512, 729, 1234.

18. List every multiple of 6 between 30 and 70 (inclusive).

19. How many multiples of 3 are there from 1 to 60?

20. For each number, list every set from $\{N, W, Z\}$ it belongs to: (a) 7 (b) 0 (c) -4.

21. True or false: (a) Every natural number is an integer. (b) Every integer is a whole number. (c) 0 is a natural number.

22. Give ONE number in each category, or write 'impossible': (a) A whole number that is not a natural number. (b) An integer that is not a whole number. (c) A natural number that is not an integer.

23. A stock opened Monday at \$50. It changed each day: Mon +4, Tue -6, Wed +2, Thu -3, Fri +5. What was the closing price on Friday?

24. In a card game, positive cards add to your score and negative cards subtract. Amara drew: +5, -3, +8, -4, and a mystery card X. Her final score was 10. What is X?

25. An elevator starts at floor 0 and makes these moves: +4, -7, +3, -2, +6. (a) What floor does it end on? (b) What is the total DISTANCE (in floors) it traveled?

ANSWER KEY — Hidden Explanations

DO NOT PEEK until you have finished the test. Each answer is hidden inside a dark box. To reveal one at a time, either (1) **highlight** the box on screen with your cursor, or (2) if printing, hold the sheet up to bright light or use a highlighter to peel back and check one answer at a time. Try the whole test on your own first!

ANSWERS WITH EXPLANATIONS

Q1. From the list -6, 0, 3, -1, 5, 0.5, -0.25, list every number that belongs to $\mathbb{Z}$ (the integers).

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Q25. An elevator starts at floor 0 and makes these moves: +4, -7, +3, -2, +6. (a) What floor does it end on? (b) What is the total DISTANCE (in floors) it traveled?

SOMATH tip: If you missed a question, don't just fix the number — read the explanation and try one more problem of the same type. That's how skill sticks.

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